

Zomato Bangalore Food Market Analysis

Data Pipeline, Advanced SQL Transformations & Executive BI Dashboard

Project Overview

This initiative examines the hyper-competitive culinary landscape of Bangalore by analyzing transactional and operational data across 41,000+ active restaurants. The objective is to build an end-to-end data pipeline that maps market structures, isolates customer satisfaction metrics, and guides strategic restaurant positioning for investors.

Dataset Summary

- **Total Rows:** 41,665 records
- **Total Features:** 11 structural columns
- **Key Attributes Analyzed:** Customer engagement (Votes), Operational segments (Cuisines, Restaurant Type), Financial metrics (Approx Cost for Two), and Performance metrics (Ratings).
- **Data Sources:** Live Relational Tables engineered inside MySQL Workbench.

Data Engineering & Advanced SQL Work

Constructed a relational schema in MySQL Workbench to load raw text files, clean data formatting issues, and aggregate structural indicators:

- **Bulk Data Loading:** Created structured targets and executed large bulk data injections directly from web-scraped source files using optimized file system delimiters.
- **Regional Density Analysis:** Aggregated location frequencies and order types to isolate dominant distribution clusters across key business areas.
- **Performance Matrix Formulation:** Parsed text ratings and extracted clean numeric data to separate high-performing local destinations from over-hyped outlets using custom multi-variable case statements.
- **Cuisine Hype Mapping:** Utilized advanced string array parsing and unnesting mechanisms to evaluate the exact volume impact of major cuisine variations.
- **Production Database View:** Consolidated all cleaned tables, custom fields, and optimized variables into a production-ready database view titled `v\_market\_analysis\_base` for direct, secure BI consumption.

```
-- Algorithmic Customer Segmentation (Hidden Gems vs Over-Hype)
WITH Metrics AS (
  SELECT name,
         CAST(NULLIF(REPLACE(rate, '/5', ''), 'NEW') AS NUMERIC) as r,
         MAX(votes) as v
  FROM zomato_raw
  WHERE rate IS NOT NULL AND rate != 'NEW' AND rate != '-'
  GROUP BY name, rate
)
```

```

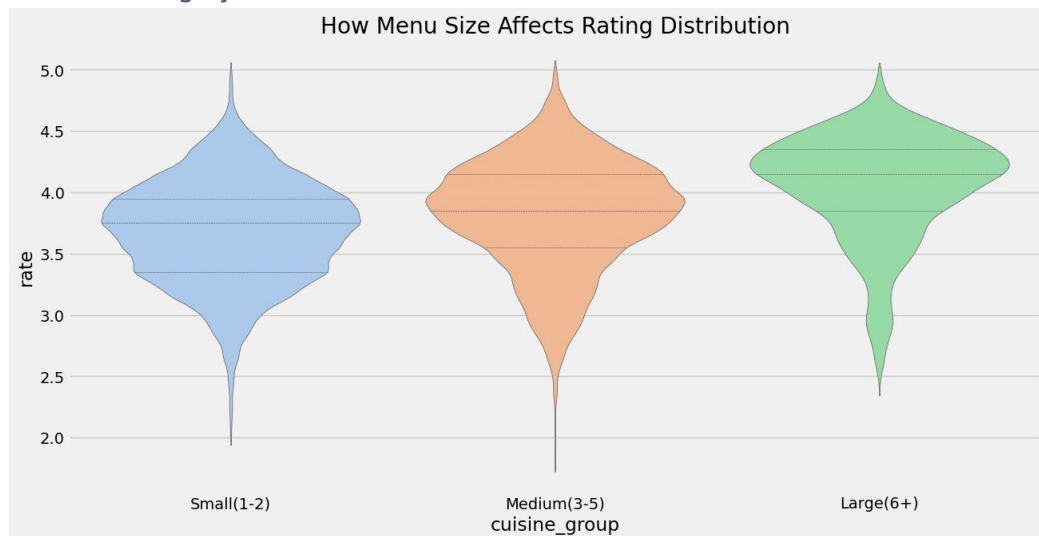
SELECT name, r as rating, v as votes,
CASE
  WHEN v > 1000 AND r < 3.8 THEN 'High Hype / Low Satisfaction'
  WHEN v < 100 AND r > 4.2 THEN 'Hidden Gem'
  ELSE 'Standard'
END as market_perception
FROM Metrics;

```

Exploratory Data Analysis (EDA) & Feature Engineering using Python

Utilized Pandas and SQLAlchemy within a Jupyter Notebook environment to handle granular data cleaning and isolate metric distributions:

- **Live Database Pipeline:** Connected directly to the live relational server to load and process structural records seamlessly without manual file extracts.
- **Rigorous Text Scrubbing:** Cleaned irregularities in text fields, stripped non-numeric elements, and standardized values for costs and consumer reviews into mathematical formats.
- **Menu Diversity Metrics:** Computed the exact number of cuisines per restaurant using comma-separation patterns to measure menu scope.
- **Operational Menu Bins:** Segmented menu counts into clean categorical brackets: Small (1-2), Medium (3-5), and Large (6+).
- **Financial Bracket Tiering:** Mapped custom monetary limits to divide outlets into 4 clear financial segments: Budget (≤ 300), Mid-Range (301-700), Premium (701-1500), and Luxury (> 1500).
- **Statistical Variance Analysis:** Generated advanced violin distribution plots to evaluate rating variances across menu sizes, revealing that tighter, focused menus deliver highly stable customer scores.



Python Violin Plot - Menu Size vs. Rating Stability Distribution

Power BI Executive Dashboard Architecture

Developed an enterprise semantic layer configured for quick strategic analysis, emphasizing data density and balanced horizontal alignment:

1. Top High-Impact Tactical KPI Strip

- **Total Active Restaurants:** Displays total active industry outlets to track absolute market scale.
- **Market Average Rating:** Establishes a baseline benchmark for customer satisfaction (3.70 / 5.00).
- **Average Cost for Two:** Tracks entry-level financial requirements for consumer dining pairs.
- **Digital Ordering Penetration (%):** Monitors online delivery adoption across major city zones.
- **Total Engagement Volume (Votes):** Aggregates overall customer feedback volume to track customer engagement.

2. Power BI Visualizations Breakdown

Visual Name / Context	Core Fields Used	Strategic Business Value
Market Density by Budget Segment	X: budget_segment Y: Count of Restaurants	Identifies volume clusters to locate price point saturations.
The Volume vs. Quality Sweet Spot	X: budget_segment Column Y: Sum of Votes Line Y: Average Rate	Pinpoints the intersection where high volume meets top user appreciation.
Strategic Market Positioning Tiers	X: Count of Restaurants Y: market_perception	Highlights market composition across Hidden Gems and High Hype clusters.
Operational Formats & Density	X: Count of Restaurants Y: rest_type	Profiles structural layout preferences (e.g., Quick Service vs. Cafes).
Regional Rating & Price Point Matrix	Rows: location Columns: budget_segment Values: Avg Rate	Heatmap displaying geographical gaps where specific budget tiers perform best.



Power BI Executive Dashboard - Full Overview Canvas Screen

Strategic Business Recommendations

- Target the Financial Sweet Spot:** Focus capital allocation within the 300 Rs to 1000 Rs segment (Mid-Range to early Premium), where customer engagement spikes sharply alongside stable ratings.
- Optimize Menu Scope Diversity:** Enforce a strict 'Medium (3-5)' cuisine count limit. Expansive menus introduce operational complexity and consistency challenges, leading to higher rating variance.
- Digital Acquisition Roadmap:** Instead of funding expensive, asset-heavy new launches, acquire existing high-rating, low-visibility outlets ('Hidden Gems') and scale them through digital marketing and delivery optimizations.